Robot Kinematics and Dynamics report

The designed manipulator is a 4-degree-of-freedom (4-DOF) selective compliance robot intended for pick-and-place tasks on a conveyor belt. The combination of two revolute joints (RR), one prismatic joint (P), and a final revolute joint (R) allow positioning in 3D space while maintaining orientation. The prismatic joint provides vertical lift, and the offset between links ensures a larger workspace without self-collision.

According to assignment file, first we design our **RRPR** robot:

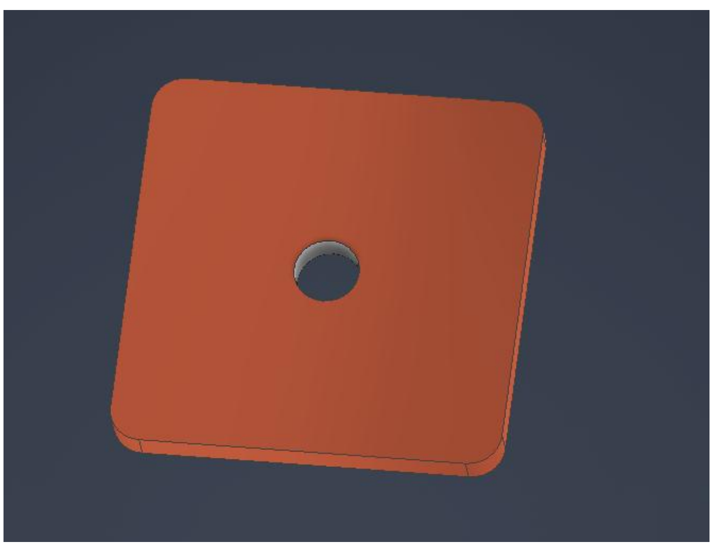
1. CAD Design (Autodesk Inventor)

- **Base (orange)**: Fixed to ground.
- **Link 1 (brown)**: Revolute joint (θ_1) about vertical axis.
- **Link 2 (yellow)**: Revolute joint (θ_2) about horizontal axis.
- **Link 3 (red)**: Prismatic joint (d_3) with an offset (non-zero distance between joint axes).
- **Link 4 (light blue)**: End-effector, revolute joint (θ_4) about vertical axis.

Parts:

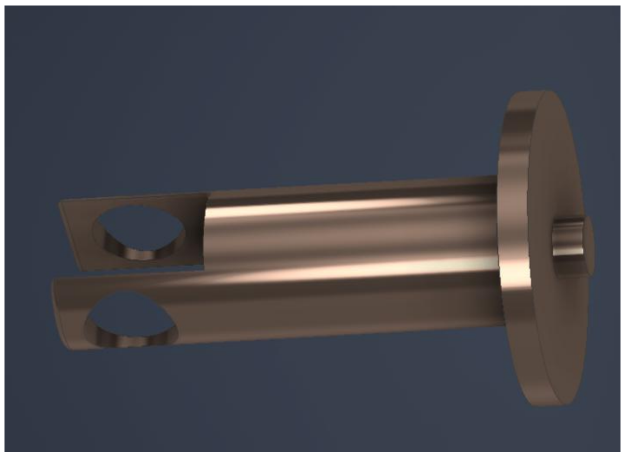
1. Ground base

As we said we have a 600 mm * 600 mm square which is fixed to the ground and it is stable and does not move. The depth of this part is 50 mm. I designed a circle as a hole in middle of this square with 100 mm diameter for connecting part 1 to that. I also added fillets for better view.



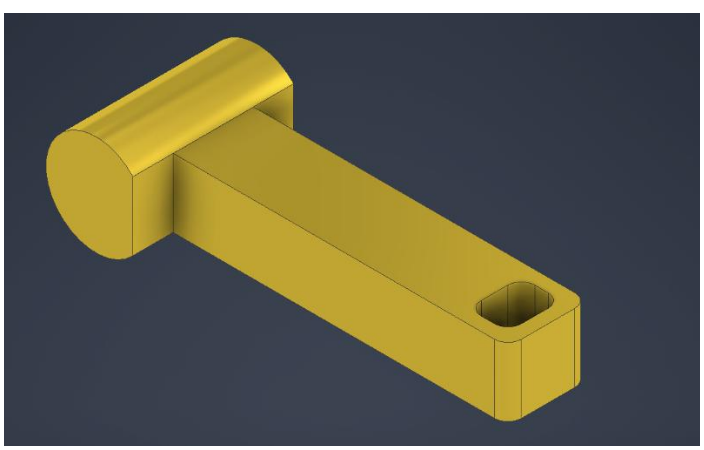
2. Part 1 (first revolute link)

Now we should have a cylinder attached to main part one to fit in the base part hole. So, the diameter of the hole is 100 mm again with 50 mm depth. The diameter of the base of this part is 500 mm. As it is shown in the picture below, there is a cylinder with 240 mm diameter and 750 mm height. This part 1 can rotate around itself (z axis). In the other side of this cylinder there is a empty space for moving next part freely and to circle hole with 150 mm diameter for attaching part 2. I also added fillets for better view. (l_1 = 650 mm)



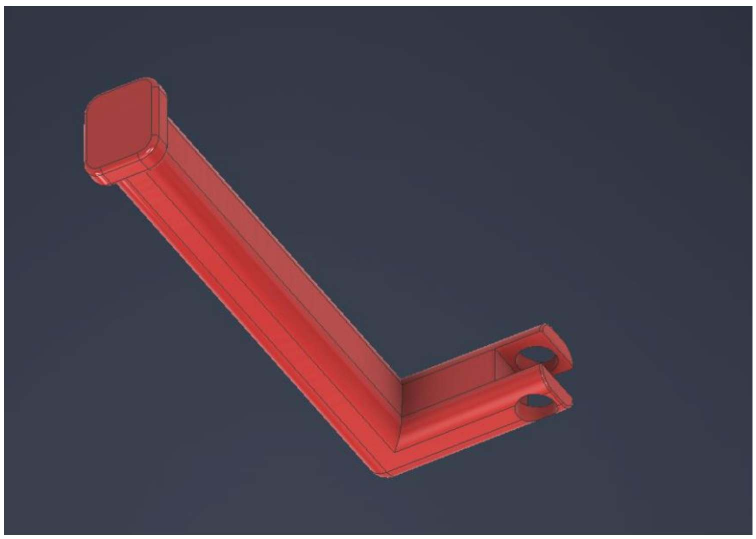
3. Part 2 (second revolute link)

In this part I made another horizontal cylinder for attaching to previous holes with same size (150 mm diameter again and the length of this cylinder is 240 mm). I removed a rectangular cube of this cylinder to attach the other part easier and prettier. Then I made a simple rectangular cube with 500 mm length (l_2), 120 mm width, (100 mm depth). At end of this part, there is 80 mm *60 mm rectangular hole (with 2 mm fillet in each side for being more attractive), for connecting the third part. So, this part can move freely. This is another rotational part.



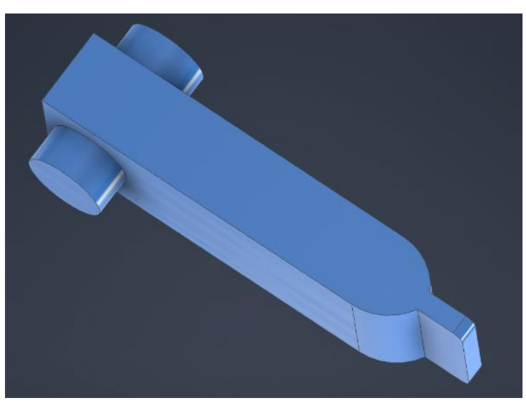
4. Part 3 (prismatic link)

This part is prismatic. For fitting in previous hole, the size of the rectangular cube with 500 mm length, should be 60 mm * 80 mm again. In begin of this part I designed a bigger rectangular which is 100 mm * 80 mm and 25 mm depth and the reason is this part does not pass the hole and stuck there. I also added fillets (this time around 5mm for the whole part) for better view. At other side, we have an offset (one non-zero offset between two consecutive link 4 & 5), which makes this part like a L shape, the lower part has 190 mm length (l_{32}), and like part 1, there are 2 holes with 40 mm diameter. There are a few 2 mm fillets for better view in the last part which is not affecting the robot application.



5. Part 4 (last revolute link)

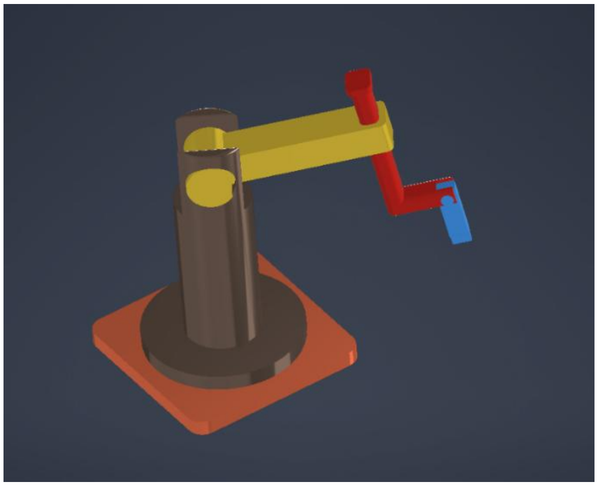
This is a revolute part again same as part 2 we put 2 pins for locking in the previous part holes. The distance of the two holes were 80 mm. So, here the length of the two pins will be 80 mm too. There is another rectangular cube with 40 mm width, 220 mm length (l_4), and 60 mm depth. End effector of this part designed pointy for being more beautiful and special functions.



Assembly part

After opening inventor, we design ground part (orange) which is fixed plus 4 parts. Part 1 (brown) and part 2 (yellow) are rotational. Part 3 (red) is prismatic with an offset and part 4 (light blue) is the last part (end effector).

In this assembly part, I limited the movement of prismatic part to not separate from the robot during the movement.

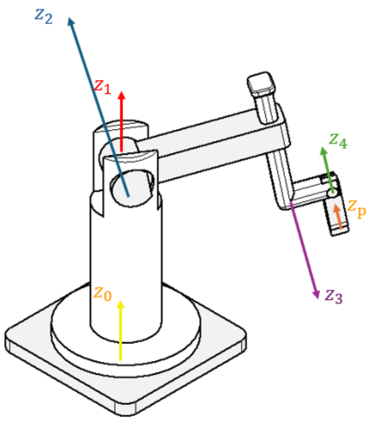


At the end we modify the initial position and export it for Adams PD controller design session.

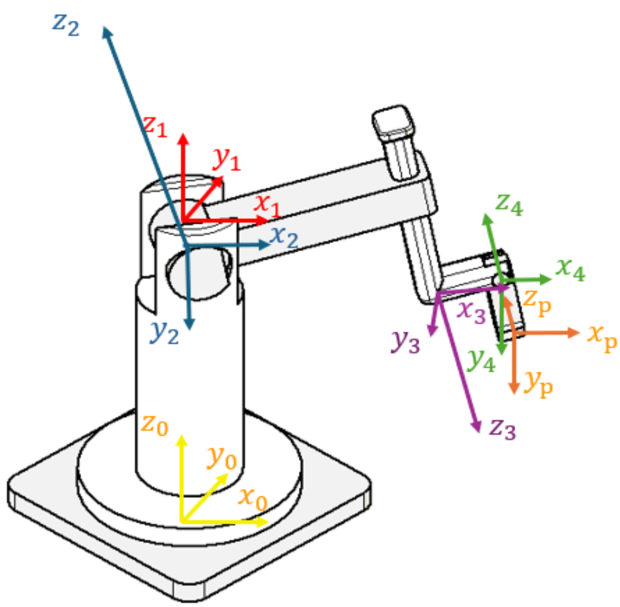
2. Forward Kinematics

2.1 DH parameters

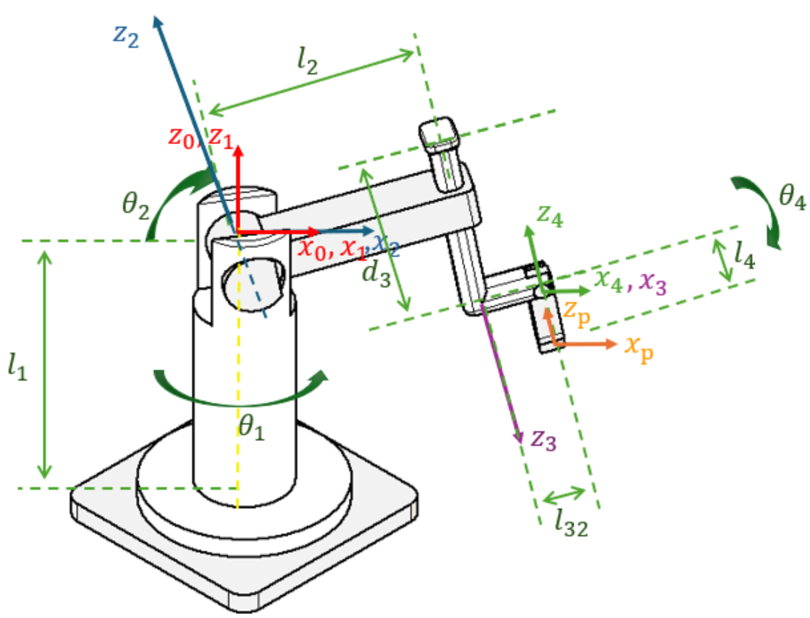
Find the DH parameters with the corresponding coordinate frames assigned to the base, the joints, and the end effector. Start to draw Z vectors:



Then we can add x vectors and y vectors:



For better view we can determine the dimensions and combine some vectores:



Now we can determine the DH table:

$$DH\ table$$

	α_{i-1}	a_{i-1}	d_i	θ_i
$i = 1$	0	0	l_1	$\theta_1(t)$
$i = 2$	$-\frac{\pi}{2}$	0	0	$\theta_2(t)$
$i = 3$	$-\frac{\pi}{2}$	l_2	$d_3(t)$	0
$i = 4$	$\frac{\pi}{2}$	l_{32}	0	$\theta_4(t)$

2.2 Homogeneous Transformation Matrices

Now, compute different homogeneous transformation matrices and solve the forward kinematics problems.

Each transformation matrix from frame {i-1} to {i} is:

$$T_{i-1}^{i} = \begin{bmatrix} \cos\theta_i & -\sin\theta_i\cos\alpha_{i-1} & \sin\theta_i\sin\alpha_{i-1} & a_{i-1}\cos\theta_i \\ \sin\theta_i & \cos\theta_i\cos\alpha_{i-1} & -\cos\theta_i\sin\alpha_{i-1} & a_{i-1}\sin\theta_i \\ 0 & \sin\alpha_{i-1} & \cos\alpha_{i-1} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

After importing this DH table in MATLAB, next step is:

After defining DH table, now we can calculate the Homogeneous Transformation Matrices with MATLAB help for each frame (use HTM file and RRPR.mlx):

T01 =

$$\begin{pmatrix} \cos(\theta_1) & -\sin(\theta_1) & 0 & 0 \\ \sin(\theta_1) & \cos(\theta_1) & 0 & 0 \\ 0 & 0 & 1 & l_1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

T12 =

$$\begin{pmatrix} \cos(\theta_2) & -\sin(\theta_2) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\sin(\theta_2) & -\cos(\theta_2) & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

T23 =

$$\begin{pmatrix} 1 & 0 & 0 & l_2 \\ 0 & 0 & 1 & d_3 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

T34 =

$$\begin{pmatrix} \cos(\theta_4) & -\sin(\theta_4) & 0 & l_{32} \\ 0 & 0 & -1 & 0 \\ \sin(\theta_4) & \cos(\theta_4) & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

2.3 End-Effector Pose T04

T04 is achieved by multiplication of T01*T12*T23*T34 :

T04 =

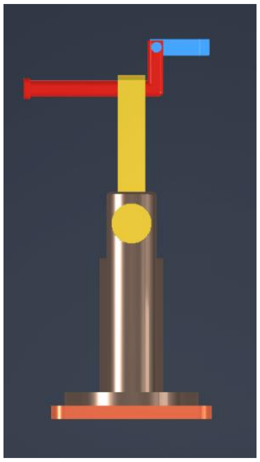
$$\begin{pmatrix} \cos(\theta_2+\theta_4)\cos(\theta_1) & -\sin(\theta_2+\theta_4)\cos(\theta_1) & -\sin(\theta_1) & \cos(\theta_1)\ (l_2\cos(\theta_2)+l_{32}\cos(\theta_2)-d_3\sin(\theta_2)) \\ \cos(\theta_2+\theta_4)\sin(\theta_1) & -\sin(\theta_2+\theta_4)\sin(\theta_1) & \cos(\theta_1) & \sin(\theta_1)\ (l_2\cos(\theta_2)+l_{32}\cos(\theta_2)-d_3\sin(\theta_2)) \\ -\sin(\theta_2+\theta_4) & -\cos(\theta_2+\theta_4) & 0 & l_1-d_3\cos(\theta_2)-l_2\sin(\theta_2)-l_{32}\sin(\theta_2) \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

T0P =

$$\begin{pmatrix} \cos(\theta_2+\theta_4)\cos(\theta_1) & -\sin(\theta_2+\theta_4)\cos(\theta_1) & -\sin(\theta_1) & \cos(\theta_1)\ (l_2\cos(\theta_2)+l_{32}\cos(\theta_2)-d_3\sin(\theta_2))-l_4\sin(\theta_2+\theta_4)\cos(\theta_1) \\ \cos(\theta_2+\theta_4)\sin(\theta_1) & -\sin(\theta_2+\theta_4)\sin(\theta_1) & \cos(\theta_1) & \sin(\theta_1)\ (l_2\cos(\theta_2)+l_{32}\cos(\theta_2)-d_3\sin(\theta_2))-l_4\sin(\theta_2+\theta_4)\sin(\theta_1) \\ -\sin(\theta_2+\theta_4) & -\cos(\theta_2+\theta_4) & 0 & l_1-l_4\cos(\theta_2+\theta_4)-d_3\cos(\theta_2)-l_2\sin(\theta_2)-l_{32}\sin(\theta_2) \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

T0P is also can be calculated by multiplying T04 in $\begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & l_4 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$ (because in this last movement robot just move for $+l_4$ in y axis).

We can test our robot by finding initial position and validating it:



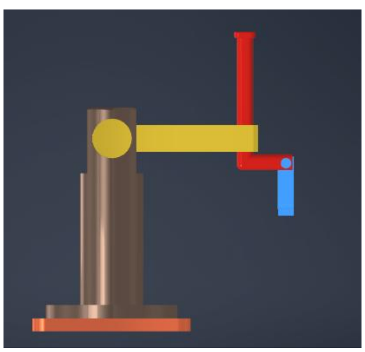
test1 =

$$\begin{pmatrix} 0 & 1 & 0 & l_4 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & l_1 + l_2 + l_{32} \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

In z axis we have l_1 (brown part) plus l_2 (yellow part) plus l_{32} (the off set of red part) and for x axis we have l_4 (light blue part) and for y axis we don't have any vector that physically makes sense. (There is a gap before starting l_4 because of limitation of movement of prismatic part but for easier calculation we consider l_4 from center of base robot to end of the blue part).

3. Inverse kinematics:

We have another test for validation:



test2 =

$$\begin{pmatrix} 1 & 0 & 0 & l_2 + l_{32} \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & l_1 - l_4 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

In this test, we have l_1 (brown part) and on opposite side is l_4 (blue part) in Z axis, which means l_1-l_4 is the transformation in this vector. (there is a gap before starting l_4 because of limitation of movement of prismatic part but for easier calculation we consider l_4 from center of yellow circle part to end of the blue part). In X axis also we have l_2 and l_{32} on the same side of positive X axis which means l_2+l_{32} is the transformation in this vector.

Note: this test is for initial position of robot too:

$$\begin{pmatrix} 1 & 0 & 0 & l_2 + l_{32} \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & l_1 - l_4 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

Forward kinematics by evaluating the end-effector pose for selected joint configurations with clear physical meaning validate and results are consistent with the expected geometry of the manipulator are verified.

$$
{}^{0}_{P}T = \begin{bmatrix} r_{11} & r_{12} & r_{13} & X_p \\ r_{21} & r_{22} & r_{23} & Y_p \\ r_{31} & r_{32} & 0 & Z_p \\ 0 & 0 & 0 & 1 \end{bmatrix} =
$$

T0P =

$$
\begin{pmatrix} \cos(\theta_2+\theta_4)\cos(\theta_1) & -\sin(\theta_2+\theta_4)\cos(\theta_1) & -\sin(\theta_1) & \cos(\theta_1)\,(l_2\cos(\theta_2)+l_{32}\cos(\theta_2)-d_3\sin(\theta_2))-l_4\sin(\theta_2+\theta_4)\cos(\theta_1) \\ \cos(\theta_2+\theta_4)\sin(\theta_1) & -\sin(\theta_2+\theta_4)\sin(\theta_1) & \cos(\theta_1) & \sin(\theta_1)\,(l_2\cos(\theta_2)+l_{32}\cos(\theta_2)-d_3\sin(\theta_2))-l_4\sin(\theta_2+\theta_4)\sin(\theta_1) \\ -\sin(\theta_2+\theta_4) & -\cos(\theta_2+\theta_4) & 0 & l_1-l_4\cos(\theta_2+\theta_4)-d_3\cos(\theta_2)-l_2\sin(\theta_2)-l_{32}\sin(\theta_2) \\ 0 & 0 & 0 & 1 \end{pmatrix}
$$

For T0P calculating it by hand is like:

$$
{}^{0}_{P}T = \begin{bmatrix} r_{11} & r_{12} & r_{13} & X_p \\ r_{21} & r_{22} & r_{23} & Y_p \\ r_{31} & r_{32} & 0 & Z_p \\ 0 & 0 & 0 & 1 \end{bmatrix} =
$$

T0P =

$$
\begin{pmatrix} \cos(\theta_2+\theta_4)\cos(\theta_1) & -\sin(\theta_2+\theta_4)\cos(\theta_1) & -\sin(\theta_1) & \cos(\theta_1)\,(l_2\cos(\theta_2)+l_{32}\cos(\theta_2)-d_3\sin(\theta_2))-l_4\sin(\theta_2+\theta_4)\cos(\theta_1) \\ \cos(\theta_2+\theta_4)\sin(\theta_1) & -\sin(\theta_2+\theta_4)\sin(\theta_1) & \cos(\theta_1) & \sin(\theta_1)\,(l_2\cos(\theta_2)+l_{32}\cos(\theta_2)-d_3\sin(\theta_2))-l_4\sin(\theta_2+\theta_4)\sin(\theta_1) \\ -\sin(\theta_2+\theta_4) & -\cos(\theta_2+\theta_4) & 0 & l_1-l_4\cos(\theta_2+\theta_4)-d_3\cos(\theta_2)-l_2\sin(\theta_2)-l_{32}\sin(\theta_2) \\ 0 & 0 & 0 & 1 \end{pmatrix}
$$

So according to that we can find first θ_1:

$$
\left.\begin{aligned} r_{13} &= -Sin\,(\theta_1) \\ r_{23} &= \ \ Cos\,(\theta_1) \end{aligned}\right\} \Rightarrow \boxed{\theta_1 = atan2\,(-r_{13}, r_{23})}
$$

$$r_{31} = -Sin\,(\theta_2+\theta_4)$$

$$r_{32} = -Cos\,(\theta_2+\theta_4)$$

$$Y_p = \sin(\theta_1)\,(l_2\cos(\theta_2)+l_{32}\cos(\theta_2)-d_3\sin(\theta_2))-l_4\sin(\theta_2+\theta_4)\sin(\theta_1)\ ,\ \frac{Y_p}{Sin\,(\theta_1)} = Cos\,(\theta_2)(l_2+l_{32})-d_3Sin\,(\theta_2)-l_4Sin\,(\theta_2+\theta_4)$$

$$\frac{Y_p}{-r_{13}} - l_4r_{31} = Cos\,(\theta_2)(l_2+l_{32})-d_3Sin\,(\theta_2)$$

$$Z_p = l_1-l_4\cos(\theta_2+\theta_4)-d_3\cos(\theta_2)-l_2\sin(\theta_2)-l_{32}\sin(\theta_2)$$

$$Z_p - l_1 + l_4Cos\,(\theta_2+\theta_4) = -d_3Cos(\theta_2)-(l_2+l_{32})Sin(\theta_2)$$

$$d_3Cos(\theta_2) = l_1 + l_4r_{32} - Z_p - (l_2+l_{32})Sin(\theta_2) \Rightarrow \boxed{d_3 = \frac{l_1+l_4r_{32}-Z_p}{Cos(\theta_2)} - \frac{(l_2+l_{32})Sin(\theta_2)}{Cos(\theta_2)}}$$

$$X_p = \cos(\theta_1)\,(l_2\cos(\theta_2)+l_{32}\cos(\theta_2)-d_3\sin(\theta_2))-l_4\sin(\theta_2+\theta_4)\cos(\theta_1)$$

$$\frac{X_p}{Cos\,(\theta_1)} = Cos\,(\theta_2)(l_2+l_{32})-d_3Sin\,(\theta_2)-l_4Sin\,(\theta_2+\theta_4)$$

$$\boxed{\frac{X_{\mathrm{p}}}{r_{23}} - l_4 r_{31} = Cos\,(\theta_2)(l_2 + l_{32}) - d_3 Sin\,(\theta_2)}$$

Now we can replace d_3 in this equation from Z_p equation, we can calculate θ_2.

$$\boxed{\theta_2 = \operatorname{atan2}\left(l_1 + l_4 r_{32} - Z_p, \frac{X_p}{r_{23}} - l_4 r_{31}\right) \pm \operatorname{atan2}\left(\sqrt{\left(\frac{X_p}{r_{23}} - l_4 r_{31}\right)^2 + (l_1 + l_4 r_{32} - Z_p)^2 - (l_2 + l_{32})^2}, l_2 + l_{32}\right)}$$

We have θ_2, we can calculate θ_4 easily by using $r_{31} = -Sin\,(\theta_2 + \theta_4)$ equation:

$$\boxed{\theta_4 = \arcsin(-r_{31}) - \theta_2}$$

Calculate the value of d_3 now by replacing the θ_2 in the $d_3 = \frac{l_1 + l_4 r_{32} - Z_{\mathrm{p}}}{Cos(\theta_2)} - \frac{(l_2 + l_{32})Sin(\theta_2)}{Cos(\theta_2)}$ equation:

$$\boxed{d_3 = \pm\sqrt{\left(\frac{X_p}{r_{23}} - l_4 r_{31}\right)^2 + (l_1 + l_4 r_{32} - Z_p)^2 - (l_2 + l_{32})^2}}$$

Note: by using trigonometric relations, we could first calculate d_3 and then find θ_2 and θ_4.

Test: now for evaluating our T0P, we put $\theta_1, \theta_2, \theta_4 = 0$ and consider $d_3 = 0$ just for easier calculations. In this case, all the cos() will be = 1 and all the sin() will be = 0. (d_3 can be any arbitrary number >0 & < 450 mm).

T0P =

$$\begin{pmatrix} \cos(\theta_2+\theta_4)\cos(\theta_1) & -\sin(\theta_2+\theta_4)\cos(\theta_1) & -\sin(\theta_1) & \cos(\theta_1)\,(l_2\cos(\theta_2)+l_{32}\cos(\theta_2)-d_3\sin(\theta_2))-l_4\sin(\theta_2+\theta_4)\cos(\theta_1) \\ \cos(\theta_2+\theta_4)\sin(\theta_1) & -\sin(\theta_2+\theta_4)\sin(\theta_1) & \cos(\theta_1) & \sin(\theta_1)\,(l_2\cos(\theta_2)+l_{32}\cos(\theta_2)-d_3\sin(\theta_2))-l_4\sin(\theta_2+\theta_4)\sin(\theta_1) \\ -\sin(\theta_2+\theta_4) & -\cos(\theta_2+\theta_4) & 0 & l_1-l_4\cos(\theta_2+\theta_4)-d_3\cos(\theta_2)-l_2\sin(\theta_2)-l_{32}\sin(\theta_2) \\ 0 & 0 & 0 & 1 \end{pmatrix} = \begin{bmatrix} 1 & 0 & 0 & l_2+l_{32} \\ 0 & 0 & 1 & 0 \\ 0 & -1 & 0 & l_1-l_4+\cancelto{0}{d_3} \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

The result will be exactly as test 2. We can put any number for d_3.

4. <u>Jacobian</u>

$$\boxed{\begin{bmatrix} \vec{V}_P \\ \vec{w}_n \end{bmatrix} = \begin{bmatrix} \vec{z}_1 \times \overrightarrow{O_1P} & \underbrace{\begin{matrix} \vec{z}_2 \times \overrightarrow{O_2P} \\ \vec{z}_2 \end{matrix}}_{\text{Revolute Joint}} & \begin{matrix} \dots \\ \dots \end{matrix} & \underbrace{\begin{matrix} \vec{z}_i \\ 0 \end{matrix}}_{\text{Prismatic Joint}} & \begin{matrix} \dots \\ \dots \end{matrix} & \vec{z}_n \times \overrightarrow{O_nP} \\ \vec{z}_1 & & & & & \vec{z}_n \end{bmatrix} \begin{bmatrix} \dot{\theta}_1 \\ \theta_2 \\ \dot{d}_i \\ \dot{\theta}_n \end{bmatrix}}$$

- **Methodology & Key Steps**

Since the robot has 4 degrees of freedom (RRPR), the Jacobian matrix will be a 4x6 matrix with 4 columns (each column belongs to a joint). Each Jacobian column J_i depends on whether the joint is Revolute or Prismatic:

Revolute: $$J_i = \begin{bmatrix} z_i \times (o_n - o_i) \\ z_i \end{bmatrix}$$ Prismatic: $$J_i = \begin{bmatrix} z_i \\ 0_{3\times1} \end{bmatrix}$$

Because the robot is a combination of revolute and prismatic joints, the best and most standard method is the Geometric Jacobian method. Extracting the Z axes of joints in the base frame. In the code, you have found the axes of motion angles by multiplying the rotation matrix by the vector $[0; 0; 1]^T$.

$$z_i = R_i^0 \cdot \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$$

So first calculate the Rotationals:

R01 =

$$\begin{pmatrix} \cos(\theta_1) & -\sin(\theta_1) & 0 \\ \sin(\theta_1) & \cos(\theta_1) & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

R03 =

$$\begin{pmatrix} \cos(\theta_1)\cos(\theta_2) & \sin(\theta_1) & -\cos(\theta_1)\sin(\theta_2) \\ \cos(\theta_2)\sin(\theta_1) & -\cos(\theta_1) & -\sin(\theta_1)\sin(\theta_2) \\ -\sin(\theta_2) & 0 & -\cos(\theta_2) \end{pmatrix}$$

R02 =

$$\begin{pmatrix} \cos(\theta_1)\cos(\theta_2) & -\cos(\theta_1)\sin(\theta_2) & -\sin(\theta_1) \\ \cos(\theta_2)\sin(\theta_1) & -\sin(\theta_1)\sin(\theta_2) & \cos(\theta_1) \\ -\sin(\theta_2) & -\cos(\theta_2) & 0 \end{pmatrix}$$

R04 =

$$\begin{pmatrix} \cos(\theta_2+\theta_4)\cos(\theta_1) & -\sin(\theta_2+\theta_4)\cos(\theta_1) & -\sin(\theta_1) \\ \cos(\theta_2+\theta_4)\sin(\theta_1) & -\sin(\theta_2+\theta_4)\sin(\theta_1) & \cos(\theta_1) \\ -\sin(\theta_2+\theta_4) & -\cos(\theta_2+\theta_4) & 0 \end{pmatrix}$$

Now we calculate the z_i:

z1 =

$$\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

z2 =

$$\begin{pmatrix} -\sin(\theta_1) \\ \cos(\theta_1) \\ 0 \end{pmatrix}$$

z3 =

$$\begin{pmatrix} -\cos(\theta_1)\sin(\theta_2) \\ -\sin(\theta_1)\sin(\theta_2) \\ -\cos(\theta_2) \end{pmatrix}$$

z4 =

$$\begin{pmatrix} -\sin(\theta_1) \\ \cos(\theta_1) \\ 0 \end{pmatrix}$$

The end-effector in the base frame:

O0P =

$$
\begin{pmatrix}
\cos(\theta_1)\ (l_2\cos(\theta_2) + l_{32}\cos(\theta_2) - d_3\sin(\theta_2)) - l_4\sin(\theta_2+\theta_4)\cos(\theta_1) \\
\sin(\theta_1)\ (l_2\cos(\theta_2) + l_{32}\cos(\theta_2) - d_3\sin(\theta_2)) - l_4\sin(\theta_2+\theta_4)\sin(\theta_1) \\
l_1 - l_4\cos(\theta_2+\theta_4) - d_3\cos(\theta_2) - l_2\sin(\theta_2) - l_{32}\sin(\theta_2)
\end{pmatrix}
$$

The distance vectors from the origin of each joint frame to the end-effector position (O0P) are computed via vector subtraction:

$$O_iP = O_P^0 - O_i^0$$

O1P = O0P − O0O1:

$$
\begin{pmatrix}
\cos(\theta_1)\ (l_2\cos(\theta_2) + l_{32}\cos(\theta_2) - d_3\sin(\theta_2)) - l_4\sin(\theta_2+\theta_4)\cos(\theta_1) \\
\sin(\theta_1)\ (l_2\cos(\theta_2) + l_{32}\cos(\theta_2) - d_3\sin(\theta_2)) - l_4\sin(\theta_2+\theta_4)\sin(\theta_1) \\
-l_4\cos(\theta_2+\theta_4) - d_3\cos(\theta_2) - l_2\sin(\theta_2) - l_{32}\sin(\theta_2)
\end{pmatrix}
=
\begin{pmatrix}
\cos(\theta_1)\ (l_2\cos(\theta_2) + l_{32}\cos(\theta_2) - d_3\sin(\theta_2)) - l_4\sin(\theta_2+\theta_4)\cos(\theta_1) \\
\sin(\theta_1)\ (l_2\cos(\theta_2) + l_{32}\cos(\theta_2) - d_3\sin(\theta_2)) - l_4\sin(\theta_2+\theta_4)\sin(\theta_1) \\
l_1 - l_4\cos(\theta_2+\theta_4) - d_3\cos(\theta_2) - l_2\sin(\theta_2) - l_{32}\sin(\theta_2)
\end{pmatrix}
-
\begin{pmatrix} 0 \\ 0 \\ l_1 \end{pmatrix}
$$

For O2P we have:

O2P = O0P − O0O2:

$$
\begin{pmatrix}
\cos(\theta_1)\ (l_2\cos(\theta_2) + l_{32}\cos(\theta_2) - d_3\sin(\theta_2)) - l_4\sin(\theta_2+\theta_4)\cos(\theta_1) \\
\sin(\theta_1)\ (l_2\cos(\theta_2) + l_{32}\cos(\theta_2) - d_3\sin(\theta_2)) - l_4\sin(\theta_2+\theta_4)\sin(\theta_1) \\
-l_4\cos(\theta_2+\theta_4) - d_3\cos(\theta_2) - l_2\sin(\theta_2) - l_{32}\sin(\theta_2)
\end{pmatrix}
=
\begin{pmatrix}
\cos(\theta_1)\ (l_2\cos(\theta_2) + l_{32}\cos(\theta_2) - d_3\sin(\theta_2)) - l_4\sin(\theta_2+\theta_4)\cos(\theta_1) \\
\sin(\theta_1)\ (l_2\cos(\theta_2) + l_{32}\cos(\theta_2) - d_3\sin(\theta_2)) - l_4\sin(\theta_2+\theta_4)\sin(\theta_1) \\
l_1 - l_4\cos(\theta_2+\theta_4) - d_3\cos(\theta_2) - l_2\sin(\theta_2) - l_{32}\sin(\theta_2)
\end{pmatrix}
-
\begin{pmatrix} 0 \\ 0 \\ l_1 \end{pmatrix}
$$

For O4P we have:

O4P = O0P − O0O4:

$$
\begin{pmatrix}
-l_4\sin(\theta_2+\theta_4)\cos(\theta_1) \\
-l_4\sin(\theta_2+\theta_4)\sin(\theta_1) \\
-l_4\cos(\theta_2+\theta_4)
\end{pmatrix}
=
\begin{pmatrix}
\cos(\theta_1)\ (l_2\cos(\theta_2) + l_{32}\cos(\theta_2) - d_3\sin(\theta_2)) - l_4\sin(\theta_2+\theta_4)\cos(\theta_1) \\
\sin(\theta_1)\ (l_2\cos(\theta_2) + l_{32}\cos(\theta_2) - d_3\sin(\theta_2)) - l_4\sin(\theta_2+\theta_4)\sin(\theta_1) \\
l_1 - l_4\cos(\theta_2+\theta_4) - d_3\cos(\theta_2) - l_2\sin(\theta_2) - l_{32}\sin(\theta_2)
\end{pmatrix}
-
\begin{pmatrix}
\cos(\theta_1)\ (l_2\cos(\theta_2) + l_{32}\cos(\theta_2) - d_3\sin(\theta_2)) \\
\sin(\theta_1)\ (l_2\cos(\theta_2) + l_{32}\cos(\theta_2) - d_3\sin(\theta_2)) \\
l_1 - d_3\cos(\theta_2) - l_2\sin(\theta_2) - l_{32}\sin(\theta_2)
\end{pmatrix}
$$

The Jacobian matrix has 6 rows (the first 3 rows are the linear velocity J_v and the second 3 rows are the rotational velocity (J_w) and 4 columns (the number of joints). Calculating the Jacobian using a geometric method for 4 degrees of freedom. Given that joints 1, 2, and 4 are revolute, and joint 3 is prismatic, the columns of the Jacobian matrix are formulated as follows:

$$J = \begin{bmatrix} J_v \\ J_\omega \end{bmatrix} \qquad J = \begin{bmatrix} z_1 \times O_1P & z_2 \times O_2P & z_3 & z_4 \times O_4P \\ z_1 & z_2 & \mathbf{0}_{3\times 1} & z_4 \end{bmatrix}$$

First column (joint R)_1: $J_1 = \begin{bmatrix} z_1 \times O_1P \\ z_1 \end{bmatrix}$ Second column (joint R_2): $J_2 = \begin{bmatrix} z_2 \times O_2P \\ z_2 \end{bmatrix}$

Third column (joint P_3): $J_3 = \begin{bmatrix} z_3 \\ \mathbf{0}_{3\times 1} \end{bmatrix}$ Fourth column (joint R_4): $J_4 = \begin{bmatrix} z_4 \times O_4P \\ z_4 \end{bmatrix}$

So the main J =

$$\begin{pmatrix} l_4 \sin(\theta_2+\theta_4)\sin(\theta_1) - \sin(\theta_1)\,(l_2\cos(\theta_2) + l_{32}\cos(\theta_2) - d_3\sin(\theta_2)) & -\cos(\theta_1)\,(l_4\cos(\theta_2+\theta_4) + d_3\cos(\theta_2) + l_2\sin(\theta_2) + l_{32}\sin(\theta_2)) & -\cos(\theta_1)\sin(\theta_2) & -l_4\cos(\theta_2+\theta_4)\cos(\theta_1) \\ \cos(\theta_1)\,(l_2\cos(\theta_2) + l_{32}\cos(\theta_2) - d_3\sin(\theta_2)) - l_4\sin(\theta_2+\theta_4)\cos(\theta_1) & -\sin(\theta_1)\,(l_4\cos(\theta_2+\theta_4) + d_3\cos(\theta_2) + l_2\sin(\theta_2) + l_{32}\sin(\theta_2)) & -\sin(\theta_1)\sin(\theta_2) & -l_4\cos(\theta_2+\theta_4)\sin(\theta_1) \\ 0 & l_4\sin(\theta_2+\theta_4) - l_2\cos(\theta_2) - l_{32}\cos(\theta_2) + d_3\sin(\theta_2) & -\cos(\theta_2) & l_4\sin(\theta_2+\theta_4) \\ 0 & -\sin(\theta_1) & 0 & -\sin(\theta_1) \\ 0 & \cos(\theta_1) & 0 & \cos(\theta_1) \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

Potential Singular Configurations

Kinematic singularities of the 4-DOF RRPR manipulator occur when the Jacobian matrix loses its full rank (rank(J) < 4), reducing the mobility of the end-effector. According to MATLAB, the rank = 4. For this specific configuration, two main types of singularities are identified:

1. Boundary Singularity: This occurs when the manipulator reaches the outer or inner limits of its workspace (e.g., when the links are fully extended or completely folded). At these configurations, the manipulator loses the ability to move linearly along the direction of the extended links.
2. Alignment/Orientation Singularity: Since Joint 1 and Joint 4 both rotate about vertical axes under nominal conditions, a singularity occurs when the orientation of Joint 2 (θ_1) aligns the axis of Joint 4 parallel to the axis of Joint 1 (z_1 parallel to z_4). In this configuration, the degrees of freedom contributed by $\dot{\theta}_1$ and $\dot{\theta}_4$ become redundant, resulting in the loss of one independent rotational capability in the workspace.

4.1 Linear:

J_v =

$$\begin{pmatrix} l_4 \sin(\theta_2+\theta_4)\sin(\theta_1) - \sin(\theta_1)\,(l_2\cos(\theta_2) + l_{32}\cos(\theta_2) - d_3\sin(\theta_2)) & -\cos(\theta_1)\,(l_4\cos(\theta_2+\theta_4) + d_3\cos(\theta_2) + l_2\sin(\theta_2) + l_{32}\sin(\theta_2)) & -\cos(\theta_1)\sin(\theta_2) & -l_4\cos(\theta_2+\theta_4)\cos(\theta_1) \\ \cos(\theta_1)\,(l_2\cos(\theta_2) + l_{32}\cos(\theta_2) - d_3\sin(\theta_2)) - l_4\sin(\theta_2+\theta_4)\cos(\theta_1) & -\sin(\theta_1)\,(l_4\cos(\theta_2+\theta_4) + d_3\cos(\theta_2) + l_2\sin(\theta_2) + l_{32}\sin(\theta_2)) & -\sin(\theta_1)\sin(\theta_2) & -l_4\cos(\theta_2+\theta_4)\sin(\theta_1) \\ 0 & l_4\sin(\theta_2+\theta_4) - l_2\cos(\theta_2) - l_{32}\cos(\theta_2) + d_3\sin(\theta_2) & -\cos(\theta_2) & l_4\sin(\theta_2+\theta_4) \end{pmatrix}$$

4.2 Angular Velocity

The angular velocity of the last link of the robot (w) is obtained by summing the rotational velocities of each joint. Its general formula is as follows:

Jw =

$$\begin{pmatrix} 0 & -\sin(\theta_1) & 0 & -\sin(\theta_1) \\ 0 & \cos(\theta_1) & 0 & \cos(\theta_1) \\ 1 & 0 & 0 & 0 \end{pmatrix}$$

Since the third joint is prismatic, it does not contribute to the angular velocity of the system. The final symbolic expression is simplified using MATLAB's (simplify(J)) function to analyze singularities and workspace constraints.

and the angular velocity of the manipulator's last link:

$$\omega_4 = J_\omega \dot{q}$$

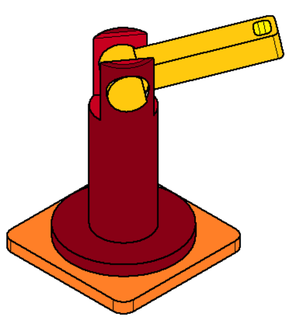
$$\omega_4 = \dot{\theta}_1 z_1 + \dot{\theta}_2 z_2 + (0) z_3 + \dot{\theta}_4 z_4 = \dot{\theta}_1 z_1 + \dot{\theta}_2 z_2 + \dot{\theta}_4 z_4$$

5. Dynamics of the Manipulator (2-DOF Case)

We assume the two joints are locked at a zero joint position, resulting in a manipulator with two degrees of freedom. The robot becomes a RR (first and second revolute) after locking part 3 and 4. Derive the centrifugal and Coriolis vector, as well as the mass matrix, while considering the two joints locked. in place at a zero joint (base part) position and two joints free to move (RR).

2D case

In this part we just need first 2 parts of the whole robot:



So, in this case the new DH table and transformation matrixes are:

DH =

$$\begin{pmatrix} 0 & 0 & l_1 & \theta_1 \\ -\frac{\pi}{2} & 0 & 0 & \theta_2 \end{pmatrix}$$

T01 =

$$\begin{pmatrix} \cos(\theta_1) & -\sin(\theta_1) & 0 & 0 \\ \sin(\theta_1) & \cos(\theta_1) & 0 & 0 \\ 0 & 0 & 1 & l_1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

T12 =

$$\begin{pmatrix} \cos(\theta_2) & -\sin(\theta_2) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ -\sin(\theta_2) & -\cos(\theta_2) & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

We can achieve T02 by multiplying T01 and T12:

T02 =

$$\begin{pmatrix} \cos(\theta_1)\cos(\theta_2) & -\cos(\theta_1)\sin(\theta_2) & -\sin(\theta_1) & 0 \\ \cos(\theta_2)\sin(\theta_1) & -\sin(\theta_1)\sin(\theta_2) & \cos(\theta_1) & 0 \\ -\sin(\theta_2) & -\cos(\theta_2) & 0 & l_1 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

The Jacobians at the COM of links are:

$$J_{G_1} = \begin{bmatrix} \vec{z_1} \times \overrightarrow{O_1G_1} & 0_{3\times1} \\ \vec{z_1} & 0_{3\times1} \end{bmatrix}, \qquad J_{G_2} = \begin{bmatrix} \vec{z_1} \times \overrightarrow{O_1G_2} & \vec{z_2} \times \overrightarrow{O_2G_2} \\ \vec{z_1} & \vec{z_2} \end{bmatrix}$$

J1 =

$$\begin{pmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 0 \\ 1 & 0 \end{pmatrix},$$

J2 =

$$\begin{pmatrix} \cos(\theta_2)\sin(\theta_1)\,(l_2 - l_{g2}) & -\cos(\theta_1)\sin(\theta_2)\,(l_2 - l_{g2}) \\ -\cos(\theta_1)\cos(\theta_2)\,(l_2 - l_{g2}) & -\sin(\theta_1)\sin(\theta_2)\,(l_2 - l_{g2}) \\ 0 & -\cos(\theta_2)\,(l_2 - l_{g2})\cos(\theta_1)^2 - \cos(\theta_2)\,(l_2 - l_{g2})\sin(\theta_1)^2 \\ 0 & -\sin(\theta_1) \\ 0 & \cos(\theta_1) \\ 1 & 0 \end{pmatrix}$$

All vectors are expressed in the base frame.

So, we should calculate new OoG1, O0O1, O0G2 for 2D case:

O0G1 = $\begin{pmatrix} 0 \\ 0 \\ l_1 - l_{g1} \end{pmatrix}$ = O0O1 = $\begin{pmatrix} 0 \\ 0 \\ l_1 \end{pmatrix}$ - O1G1 = $\begin{pmatrix} 0 \\ 0 \\ -l_{g1} \end{pmatrix}$, O1G1 = $\begin{pmatrix} 0 \\ 0 \\ -l_{g1} \end{pmatrix}$ = O0G1 = $\begin{pmatrix} 0 \\ 0 \\ l_1 - l_{g1} \end{pmatrix}$ - O0O1 = $\begin{pmatrix} 0 \\ 0 \\ l_1 \end{pmatrix}$

O1G1=O0G1-O0O1, O1G2 =O0O1-O0G2, O2G2=O0G2-O0O2

O0G2 =

$$\begin{pmatrix} \cos(\theta_1)\cos(\theta_2)\ (l_2 - l_{g2}) \\ \cos(\theta_2)\sin(\theta_1)\ (l_2 - l_{g2}) \\ l_1 - \sin(\theta_2)\ (l_2 - l_{g2}) \end{pmatrix}$$

O1G2 =

$$\begin{pmatrix} -\cos(\theta_1)\cos(\theta_2)\ (l_2 - l_{g2}) \\ -\cos(\theta_2)\sin(\theta_1)\ (l_2 - l_{g2}) \\ \sin(\theta_2)\ (l_2 - l_{g2}) \end{pmatrix}$$

O2G2 =

$$\begin{pmatrix} \cos(\theta_1)\cos(\theta_2)\ (l_2 - l_{g2}) \\ \cos(\theta_2)\sin(\theta_1)\ (l_2 - l_{g2}) \\ -\sin(\theta_2)\ (l_2 - l_{g2}) \end{pmatrix}$$

5.1. Centrifugal and Coriolis Vector

As we have these equations:

Computation of the Centrifugal and Coriolis Matrix

$$\boxed{C(\theta,\dot{\theta}) = Col\left[\sum_{i=1}^{2}\sum_{j=1}^{2}\left(\frac{\partial M_{ki}}{\partial \theta_j} - \frac{1}{2}\frac{\partial M_{ij}}{\partial \theta_k}\right)\dot{\theta}_i\dot{\theta}_j\right]}$$

Then we can express as:

$$C(\theta,\dot{\theta}) = \begin{bmatrix} \frac{1}{2}\frac{\partial M_{11}}{\partial \theta_1}\dot{\theta}_1{}^2 + \frac{\partial M_{11}}{\partial \theta_2}\dot{\theta}_1\dot{\theta}_2 + \left(\frac{\partial M_{12}}{\partial \theta_2} - \frac{1}{2}\frac{\partial M_{22}}{\partial \theta_1}\right)\dot{\theta}_2{}^2 \\ \left(\frac{\partial M_{21}}{\partial \theta_1} - \frac{1}{2}\frac{\partial M_{11}}{\partial \theta_2}\right)\dot{\theta}_1{}^2 + \frac{\partial M_{22}}{\partial \theta_1}\dot{\theta}_2\dot{\theta}_1 + \frac{1}{2}\frac{\partial M_{22}}{\partial \theta_2}\dot{\theta}_2{}^2 \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1}{2}\frac{\partial M_{11}}{\partial \theta_1} & \frac{\partial M_{12}}{\partial \theta_2} - \frac{1}{2}\frac{\partial M_{22}}{\partial \theta_1} \\ \frac{\partial M_{21}}{\partial \theta_1} - \frac{1}{2}\frac{\partial M_{11}}{\partial \theta_2} & \frac{1}{2}\frac{\partial M_{22}}{\partial \theta_2} \end{bmatrix}\begin{bmatrix} \dot{\theta}_1{}^2 \\ \dot{\theta}_2{}^2 \end{bmatrix} + \begin{bmatrix} \frac{\partial M_{11}}{\partial \theta_2} \\ \frac{\partial M_{22}}{\partial \theta_1} \end{bmatrix}\dot{\theta}_1\dot{\theta}_2$$

$$CC_1 = \begin{bmatrix} \frac{1}{2}\frac{\partial M_{11}}{\partial \theta_1} & \frac{\partial M_{12}}{\partial \theta_2} - \frac{1}{2}\frac{\partial M_{22}}{\partial \theta_1} \\ \frac{\partial M_{21}}{\partial \theta_1} - \frac{1}{2}\frac{\partial M_{11}}{\partial \theta_2} & \frac{1}{2}\frac{\partial M_{22}}{\partial \theta_2} \end{bmatrix} \qquad CC_2 = \begin{bmatrix} \frac{\partial M_{11}}{\partial \theta_2} \\ \frac{\partial M_{22}}{\partial \theta_1} \end{bmatrix}$$

CC1 =

$$\begin{pmatrix} 0 & 0 \\ \frac{\sin(2\,\theta_2)\ \left(m_2\, l_2^2 - 2\, m_2\, l_2\, l_{g2} + m_2\, l_{g2}{}^2 - I_{2,1} + I_{2,2}\right)}{2} & 0 \end{pmatrix}$$

ans =

$$\begin{pmatrix} -\sin(2\,\theta_2)\,\left(m_2\,l_2{}^2 - 2\,m_2\,l_2\,l_{g2} + m_2\,l_{g2}{}^2 - I_{2,1} + I_{2,2}\right) \\ 0 \end{pmatrix}$$

CC2 =

5.2. Mass Matrix

The extracted inertial properties of active manipulator from Autodesk Inventor using physical properties tool (I1 as a steel carbon and I2 as a steel material):

I1 =

$$\begin{pmatrix} I_{1,1} & 0 & 0 \\ 0 & I_{1,2} & 0 \\ 0 & 0 & I_{1,3} \end{pmatrix}$$

I2 =

$$\begin{pmatrix} I_{2,1} & 0 & 0 \\ 0 & I_{2,2} & 0 \\ 0 & 0 & I_{2,3} \end{pmatrix}$$

I1 =

$$\begin{pmatrix} \frac{6615854217368175}{536870912} & 0 & 0 \\ 0 & \frac{6655935248922051}{536870912} & 0 \\ 0 & 0 & \frac{4106593476034953}{1073741824} \end{pmatrix}$$

I2 =

$$\begin{pmatrix} \frac{4923698143893127}{17179869184} & 0 & 0 \\ 0 & \frac{2704019184802595}{1073741824} & 0 \\ 0 & 0 & \frac{5663150377491497}{2147483648} \end{pmatrix}$$

L_{g1} = l1 – X = 650 mm – 234.306 mm = 415.694 mm, Mass 1 = 269.864 kg

L_{g2} = l2 – X = 500 mm - 168.176 mm = 331.824 mm, Mass 2 = 75.015 kg

The Mass Matrix $M(q)$ can be expressed as:

$$\boxed{M(q) = \sum_{i=1}^{n}\left(m_i\left(J_v^i\right)^t J_v^i + \left(J_w^i\right)^t\, {}_i^0R I'_{G_i}\, {}_i^0R^t J_w^i\right)}$$

$$M = m_1(J_v^1)^t J_v^1 + (J_w^1)^t\, {}_1^0R I_{G_1}\, {}_1^0R^t J_w^1 + m_2(J_v^2)^t J_v^2 + (J_w^2)^t\, {}_2^0R I_{G_2}\, {}_2^0R^t J_w^2$$

M =

$$\begin{pmatrix} I_{1,3} + I_{2,2}\cos(\theta_2)^2 + I_{2,1}\sin(\theta_2)^2 + m_2\cos(\theta_1)^2\cos(\theta_2)^2\,(l_2 - l_{g2})^2 + m_2\cos(\theta_2)^2\sin(\theta_1)^2\,(l_2 - l_{g2})^2 & 0 \\ 0 & m_2\,l_2{}^2 - 2\,m_2\,l_2\,l_{g2} + m_2\,l_{g2}{}^2 + I_{2,3} \end{pmatrix}$$

5.3 Total Potential Energy

Under the assumption that two joints are locked in place and the manipulator has two degrees of freedom. Sum of the potential energies of the two mobile links is V and , the potential energy is:

$V = V1 + V2$

Where:

- $V1$: Potential Energy of link 1.
- $V2$: Potential Energy of link 2.

$$V1 = 9.81\, m_1\, (l_1 - l_{g1})$$

$$V2 = 9.81\, m_2\, (l_1 - \sin(\theta_2)\, (l_2 - l_{g2}))$$

$$V = 9.81\, m_1\, (l_1 - l_{g1}) + 9.81\, m_2\, (l_1 - \sin(\theta_2)\, (l_2 - l_{g2}))$$

5.4 Gravity Vector

Gravity Vector (the torque vector due to gravity terms in the equation of motion), considering that only the two joints are free to move.

$$\boxed{G(\theta) = \begin{bmatrix} \dfrac{\partial V}{\partial \theta_1} \\ \dfrac{\partial V}{\partial \theta_2} \end{bmatrix}}$$

Our Gravity Vector can be expressed as:

$$\text{Gravity} = \begin{pmatrix} 0 \\ -9.81\, m_2 \cos(\theta_2)\, (l_2 - l_{g2}) \end{pmatrix}$$

Equations of motion:

In the presence of no external forces, the general form of the equations of motion is:

$$\tau = M(\theta)\ddot{\theta} + C\left(\theta, \dot{\theta}\right) + G(\theta)$$

We define a geometric/mass constant term as follows:

$$\Delta I = I_{2,2} - I_{2,1} + m_2(l_2 - l_{g2})^2$$

So τ =

$$\begin{bmatrix} \theta_1(I_{1,3} + I_{2,1} + \Delta I\cos^2(\theta_2)) - \dot{\theta}_1\dot{\theta}_2\Delta I\sin(2\theta_2) & \theta_1(I_{1,3} + I_{2,1} + \Delta I\cos^2(\theta_2)) \\ \frac{1}{2}\dot{\theta}_1^2\Delta I\sin(2\theta_2) + \theta_2(I_{2,3} + m_2L_c^2) - M_g\cos(\theta_2) & \frac{1}{2}\dot{\theta}_1^2\Delta I\sin(2\theta_2) + \theta_2(I_{2,3} + m_2L_c^2) - M_g\cos(\theta_2) \end{bmatrix}$$

6. Control in Joint Space (PD Controller)

We are using **ADAMS**-View (two joints are free to move):

1. After exporting our assembly in the Inventor file, and importing it to ADAMS-View,
2. In order to accurately calculate physical properties and inertia, the material of all parts is set to “steel”.
3. Now we define the markers in right place and axis and use connectors part based on the vectors. We use fix connector fore base, and revolute for part 1,2 and 4 and prismatic for part 3.
4. Lock two joints so that the manipulator has two degrees of freedom.
5. Now start designing PD.:

For joint 1 we have in this case, ADAMS assigns MARKER_14 and MARKER_15, and for joint 2 we have ADAMS assigns MARKER_20 and MARKER_21.

For revolute joints we have:

$$\tau = K_p(\theta_d - \theta_m) + K_d(\dot{\theta}_d - \dot{\theta}_m) \qquad \text{Eq-1}$$

Where:

θ_d: Desired joint angle $\quad \dot{\theta}_d$: Desired angular velocity $\quad K_p$: Proportional gain

θ_m: Measured joint angle $\quad \dot{\theta}_m$: Measured angular velocity $\quad K_d$: Derivative gain

After configuring the model to accept joint torques and return the computed joint positions and velocities:

Error_Angular_Position_Joint_2 = Desired_Theta_2 - Theta_2

Error_Angular_Position_Joint_1 = Desired_Theta_1 - Theta_1

Error_Angular_Velocity_Joint_2 = Desired_Theta_2P - Theta_2P

Error_Angular_Velocity_Joint_1 = Desired_Theta_1P - Theta_1P

SFORCE_3 for first link:

*SFORCE_3 = Kp*Error_Angular_Position_Joint_1 + Kv* Error_Angular_Velocity_Joint_1*

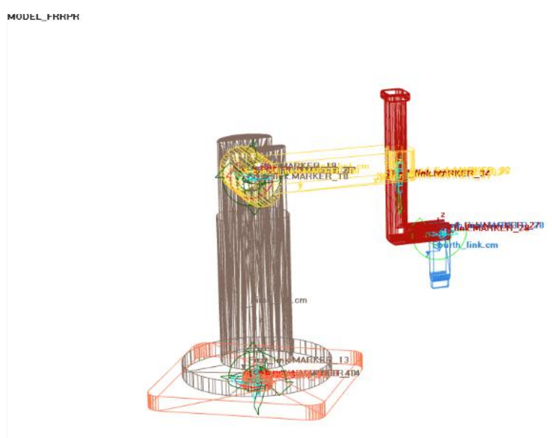
SFORCE_2 for second link:

*SFORCE_2 = Kp_2*Error_Angular_Position_Joint_2 + Kv_2* Error_Angular_Velocity_Joint_2*

Evaluate controller performance for step input:

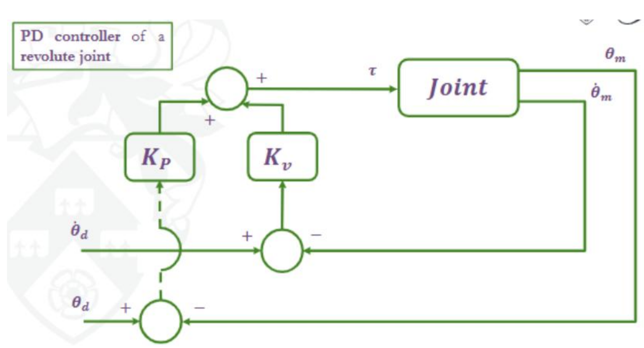
We defied theta_1, theta_2, as our measurement Desired_Theta_1 = pi/4 and Desired_Theta_2 = - pi/4 are our input. theta_1P, theta_2P measured angular velocity.

Desired_Theta_1P = 0, Desired_Theta_2P = 0 because they are the derivative of step function, which is zero.

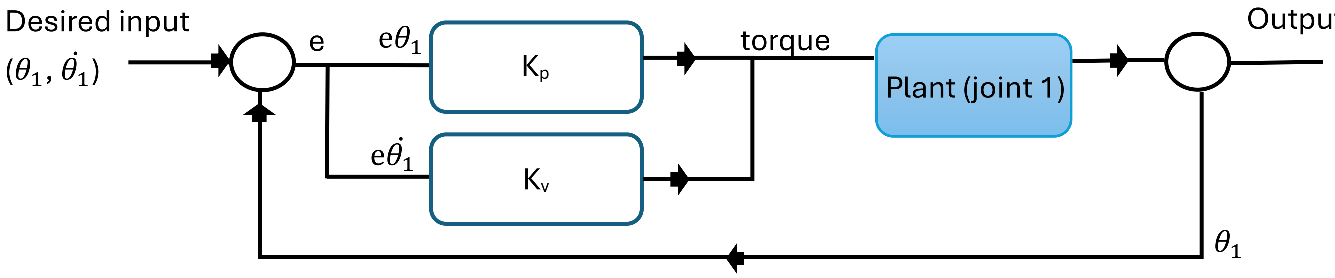


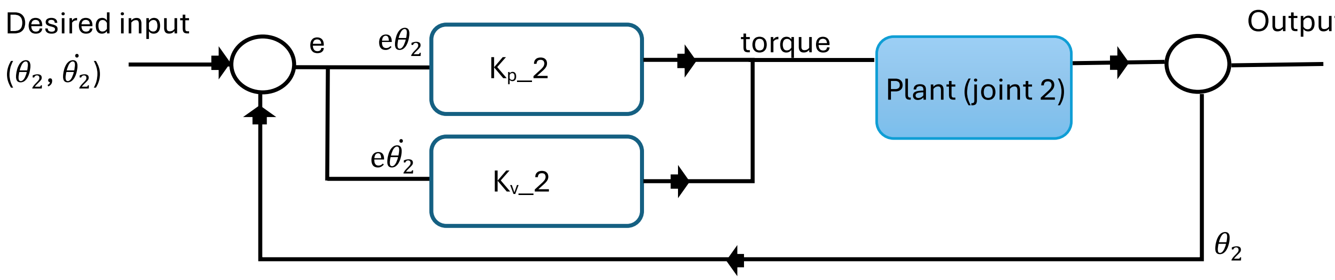
PD controller with joint position and velocity feedback:

The controller receives joint position and velocity information from the robot model and sends torque commands to drive the robot to a desired joint position.



The control law diagram:

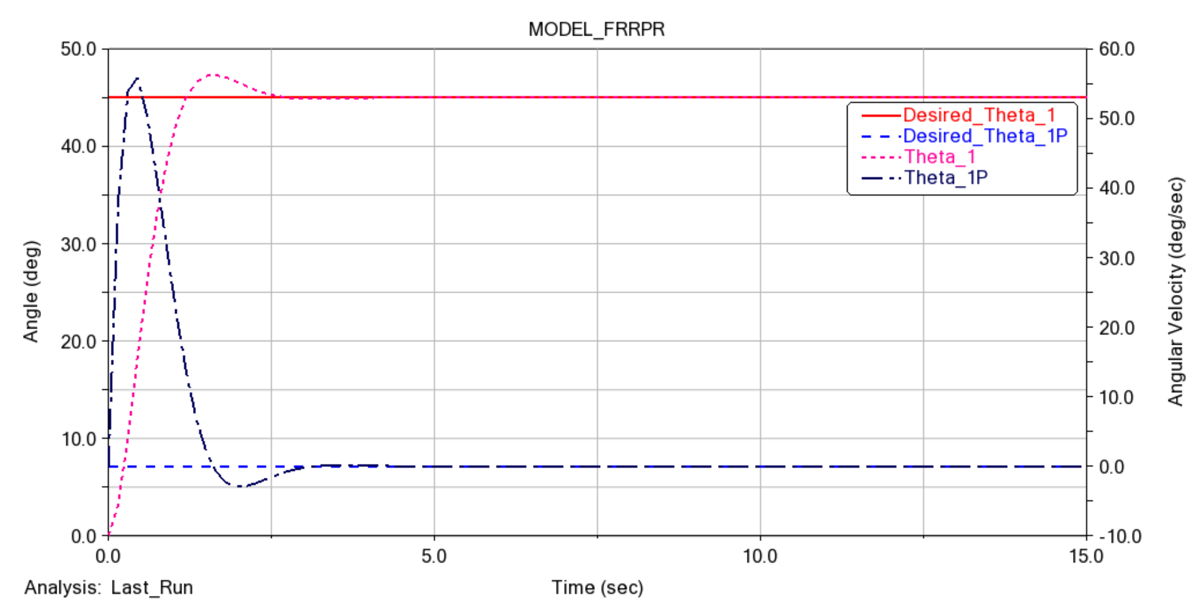




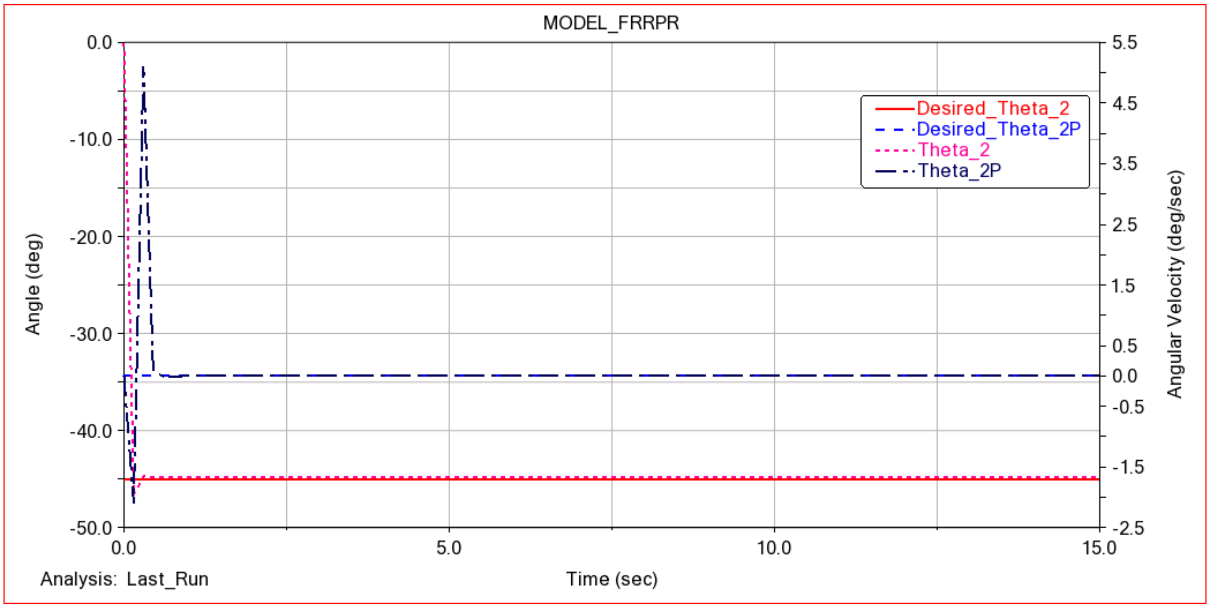
For each joint I have designed a PD controller, (kv for velocity and kp for displacement for joint one (kv_2 for velocity and kp_2 for displacement for second joint) to control better independently.

If: Kp_2 = 5.0E+07, Kv_2 = 5.0E+05, Kv = 5.0E+04, Kp = 1.0E+05

So we have below results after applying PD controller:



(for first joint)



(for second joint)